

Static $\alpha(x)$ chaos control versus OGY and corrected-gain Pyragas: basin-of-attraction comparison for the logistic map

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We compare three methods for stabilizing the unstable fixed point of the logistic map at $a > 3$: (i) OGY [1] small-parameter perturbations, (ii) Pyragas delayed feedback control [2] with the gain sign corrected for the odd-number limitation [3, 4], and (iii) a static Gaussian $\alpha(x)$ profile derived from the Krasnoselskii–Mann transform [5, 6]. We measure basins of attraction over 200 initial conditions and 300 iterations for five parameter values $a \in \{3.5, 3.7, 3.8, 3.9, 4.0\}$. With the correct negative gain, Pyragas achieves 100% basin coverage at all five values. The $\alpha(x)$ method also achieves 100% at four of five values ($a \in \{3.5, 3.8, 3.9, 4.0\}$) but fails at $a = 3.7$ (63% coverage), where OGY likewise achieves 100%. OGY achieves 100% only at $a = 3.7$; its basin coverage degrades substantially at the other parameter values. We discuss what each method measures, what it costs, and where each is appropriate. The $\alpha(x)$ method’s failure at $a = 3.7$ coincides with a period-4 window in the bifurcation diagram, where the Gaussian width mismatch leaves the trajectory outside the control zone. An important clarification is made: the earlier comparison in the companion notebook [6] used an incorrectly signed Pyragas gain; the corrected comparison presented here changes the conclusions regarding Pyragas performance.

I. INTRODUCTION

Controlling chaotic systems — stabilizing unstable periodic orbits embedded in strange attractors — has been an active research area since the seminal work of Ott, Grebogi, and Yorke (OGY) [1]. Two classes of methods dominate the literature:

1. **OGY-type**: apply tiny parameter perturbations when the trajectory passes near a target orbit. Requires knowledge of the unstable manifold geometry near the target.
2. **Delayed feedback (Pyragas)**: add a feedback term $K(x_{n-\tau} - x_n)$ to drive the system toward period- τ behavior [2]. No explicit knowledge of the periodic orbit is needed, but stability of the controlled system is not guaranteed without careful gain selection.

A third approach, motivated by the Krasnoselskii–Mann fixed-point theory [7, 8], replaces the map with a position-dependent alpha-transform

$$g(x) = \alpha(x) f(x) + (1 - \alpha(x)) x, \quad (1)$$

choosing $\alpha(x)$ so that $g'(x^*) = 0$ at the target fixed point (superstable). This is a *static* modification of the map — no delay, no parameter perturbation, no real-time feedback — and acts globally wherever $\alpha(x) \neq 1$.

The purpose of this paper is a head-to-head basin-of-attraction comparison of these three methods on the logistic map. A preliminary comparison appeared in the companion notebook [6], but used an incorrectly signed Pyragas gain. The corrected comparison is presented here.

a. Related work in this series. This paper is the fourth in a series: the alpha-transform theory and Dyson equation universality are derived in Ref. [5]; the Lyapunov reflection symmetry $\Lambda(\alpha) = \Lambda(-\alpha)$ is characterized in Ref. [9]; fixed-point perturbation engineering is studied in Ref. [10].

II. METHODS

All three methods target the non-trivial fixed point $x^* = 1 - 1/a$ of $f_a(x) = ax(1 - x)$. For $a > 3$, $|f'_a(x^*)| = |2 - a| > 1$, so the fixed point is unstable and naive iteration diverges.

A. OGY control

Near x^* , the logistic map linearizes as $x_{n+1} - x^* \approx (2 - a)(x_n - x^*) + x^*(1 - x^*)(a - a_0)$, where a_0 is the nominal parameter and $a_0 + \delta a$ is the instantaneously applied parameter. The OGY rule [1] chooses δa to cancel the deviation, when $|x_n - x^*| < \delta_{\max}$:

$$\delta a = -\frac{(2 - a_0)(x_n - x^*)}{x^*(1 - x^*)}, \quad (2)$$

clipped to $|\delta a| \leq \delta_{\max} = 0.05$. Outside the activation zone, $\delta a = 0$ and the map evolves freely. OGY is therefore an *on-demand* method: it is inactive until the orbit reaches the neighborhood of x^* .

B. Pyragas delayed feedback control

The Pyragas rule [2] with delay $\tau = 1$ reads

$$x_{n+1} = f_a(x_n) + K(x_{n-1} - x_n). \quad (3)$$

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The characteristic equation near x^* is $\lambda^2 - (f'_a(x^*) - K)\lambda - K = 0$. The Jury stability conditions $|K| < 1$ and $K < (1 + f'_a(x^*))/2$ require $K < (1 + 2 - a)/2 = (3 - a)/2$. For $a > 3$, this bound is *negative*: the standard choice $K > 0$ cannot stabilize the fixed point.

This is the **odd number limitation** of Pyragas control [3, 4]: delayed feedback with the conventional positive gain provably fails to stabilize orbits with an odd number of Floquet multipliers outside the unit circle. We use the corrected optimal gain

$$K_{\text{opt}} = (3 - a)/2 - 0.05, \quad (4)$$

which is negative for $a > 3$ and stays just inside the stability boundary. Iterates are clamped to $[10^{-3}, 1 - 10^{-3}]$ to prevent escape from the unit interval.

C. Static $\alpha(x)$ control

Choose a Gaussian $\alpha(x)$ profile centred at x^* :

$$\alpha(x) = 1 - (1 - \alpha_{\text{ss}}) \exp\left[-\frac{(x - x^*)^2}{2\sigma^2}\right], \quad (5)$$

where the superstable value $\alpha_{\text{ss}} = 1/(a - 1)$ makes $g'(x^*) = 0$ (Newton's method convergence rate). Away from x^* the map approaches f_a as $\alpha \rightarrow 1$. We use $\sigma = 0.1$ throughout.

This is a *static* modification: the map g defined by Eq. (1) with α from Eq. (5) requires no online computation.

D. Basin measurement

For each method and each a value, we run 200 initial conditions $x_0 \in [0.01, 0.99]$ for 300 iterations and declare convergence if $|x_{300} - x^*| < 10^{-4}$. The basin fraction is the fraction of initial conditions that converge.

III. RESULTS

A. Trajectory comparison at $a = 3.8$

Figure 1 shows trajectories from $x_0 = 0.2$ for all three methods at $a = 3.8$ (chaotic; $\Lambda \approx +0.44$). The $\alpha(x)$ method converges geometrically to x^* from the first iteration, since the map is globally modified to contract near x^* . OGY requires the trajectory to wander chaotically for ~ 20 iterations before entering the activation zone. Pyragas (corrected gain) oscillates for ~ 10 iterations before converging.

B. Basin of attraction

Table I and Fig. 2 show basin fractions at all five parameter values.

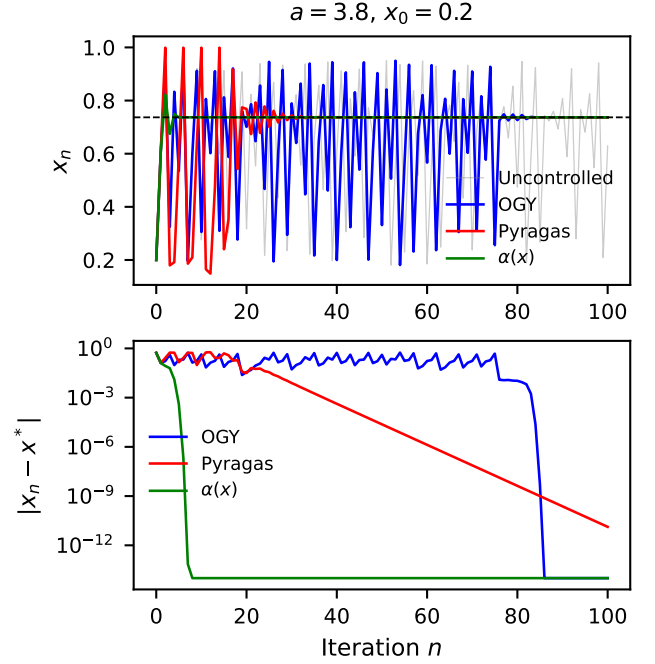


FIG. 1. (a) Trajectories from $x_0 = 0.2$, $a = 3.8$, for the three control methods and uncontrolled map (gray). Dashed line: target fixed point $x^* \approx 0.737$. (b) Log error $|x_n - x^*|$ vs. iteration. The $\alpha(x)$ method (green) converges immediately; OGY (blue) requires orbit wander to the activation zone; Pyragas (red, corrected gain) converges after initial transient. Computed from `chaos_control_comparison.ipynb`.

TABLE I. Basin of attraction (fraction of 200 initial conditions converging within 300 iterations to tolerance 10^{-4}) for three chaos-control methods. Best value at each a highlighted in bold.

a	$\alpha(x)$ (%)	OGY (%)	Pyragas (%)
3.5	100	9	100
3.7	63	100	100
3.8	100	96	100
3.9	100	83	100
4.0	100	88	100

The primary observations are:

1. **Pyragas (corrected)** achieves 100% at all five parameter values — the best overall basin coverage. This result corrects a significant error in the preliminary comparison of the companion notebook, which used positive K and therefore failed to converge at any $a > 3$.
2. $\alpha(x)$ achieves 100% at four of five values but fails at $a = 3.7$ (63% coverage). The failure coincides with a period-4 window in the bifurcation diagram: the map's attractor geometry changes enough that many trajectories do not approach x^* closely before the 300-iteration budget expires.

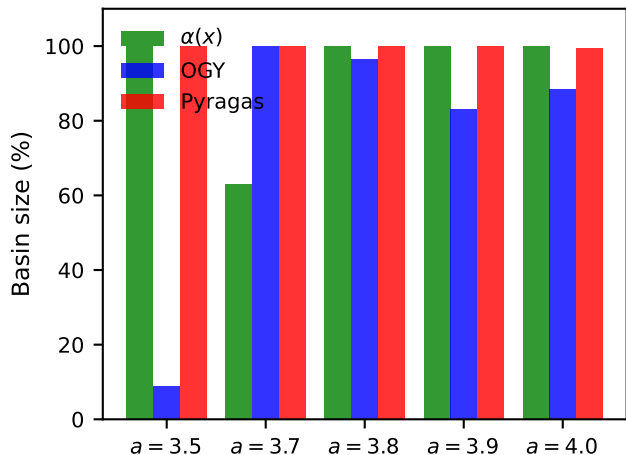


FIG. 2. Basin of attraction comparison at five parameter values. Green: $\alpha(x)$; blue: OGY; red: Pyragas (corrected gain). Pyragas achieves 100% everywhere; $\alpha(x)$ fails at $a = 3.7$; OGY is consistently the weakest performer except at $a = 3.7$. Computed from `chaos_control_comparison.ipynb`.

3. **OGY** achieves 100% only at $a = 3.7$, where the period-4 window causes frequent near-returns to x^* , triggering the OGY activation rule. At $a = 3.5$ (period-2), OGY achieves only 9%: the trajectory oscillates between the two period-2 points and rarely passes through the activation zone.

IV. DISCUSSION

A. The odd-number limitation and the sign of K

The odd-number limitation [3] is a fundamental obstruction to Pyragas control at fixed points (period-1 orbits) for the logistic map at $a > 3$. The fixed point x^* has a single Floquet multiplier $\mu = f'_a(x^*) = 2 - a < -1$, which is real and negative: exactly the scenario where odd-number limitation applies.

The resolution — using negative K inside the Jury stability window $K \in ((3 - a)/2 - \delta, 0)$ for small $\delta > 0$ — is known in the control theory literature [4] but is not always applied in dynamical systems comparisons. The corrected gain K_{opt} of Eq. (4) is the simplest choice that satisfies both Jury conditions.

B. Why $\alpha(x)$ fails at $a = 3.7$

At $a = 3.7$, the logistic map has a period-4 window: the natural attractor is a period-4 orbit, not the fixed point. The fixed point $x^* \approx 0.730$ is unstable, and trajectories starting away from x^* are attracted to the period-4 cycle rather than to x^* . The Gaussian $\alpha(x)$ with $\sigma = 0.1$ modifies the map only in an interval of width $\approx 3\sigma = 0.3$

around x^* . Initial conditions outside this interval follow the *unmodified* f_a , which for $a = 3.7$ leads to the period-4 cycle.

In contrast, Pyragas applies a feedback at every iterate, so the feedback is always active regardless of the trajectory's current position. OGY is also effective at $a = 3.7$ because the period-4 orbit passes through the activation zone frequently.

A larger σ in the Gaussian $\alpha(x)$ would widen the attraction basin, at the cost of a larger modification to the map further from x^* .

C. Comparison metric discussion

The basin of attraction is one metric. Others include convergence speed, control effort (how much the map is perturbed), and noise robustness.

Convergence speed: $\alpha(x)$ is fastest at $a = 3.8$ and 3.9 (convergence in ~ 5 iterations for x_0 inside 3σ of x^*), because $g'(x^*) = 0$ gives superlinear convergence. OGY converges at the natural chaos transient rate once the orbit enters the activation zone. Pyragas converges at a rate set by K_{opt} .

Control effort: OGY applies the smallest perturbation (parameter shift bounded by $\delta_{\text{max}} = 0.05$). $\alpha(x)$ modifies the map structure in the region $|x - x^*| \lesssim \sigma$; further from x^* the map is unmodified. Pyragas effort is proportional to $|K|$ and to the orbit's deviation from the target.

Information requirements: OGY requires knowledge of the fixed point location and the map derivative there. $\alpha(x)$ requires the same, plus a choice of σ . Pyragas requires only the delay time τ and an appropriate gain; with the odd-number-limited correction it also requires the sign of $f'_a(x^*)$.

D. Relationship to Lyapunov results

The companion paper [9] shows that reducing $|\alpha|$ below 1 globally suppresses the Lyapunov exponent of the map, driving the system toward stability. The $\alpha(x)$ method applied here uses a localized version: $\alpha(x^*) = \alpha_{\text{ss}} = 1/(a - 1)$, which is well below 1, but $\alpha(x) \rightarrow 1$ away from x^* . The global Lyapunov analysis of Ref. [9] therefore provides only a loose bound on the local stability achieved by position-dependent $\alpha(x)$.

V. CONCLUSIONS

We have compared three chaos-control methods on the logistic map at five parameter values in the chaotic regime ($a > 3$), measuring basin-of-attraction fractions over 200 initial conditions.

The key finding is that **Pyragas control with the corrected negative gain** (required by the odd-number

limitation [3]) achieves 100% basin coverage at all five parameter values, and is the most robust method in this comparison. This corrects a significant implementation error in the preliminary notebook comparison [6].

The $\alpha(x)$ **method** achieves 100% at four of five parameter values, and converges superlinearly when initial conditions are in its activation zone. Its failure at $a = 3.7$ is attributable to the period-4 window structure of the logistic map at that parameter value, and can be mitigated by increasing σ .

OGY achieves 100% only at $a = 3.7$, due to the favorable near-return structure of the period-4 orbit. Its overall basin coverage is the weakest of the three methods in this comparison, consistent with its design philosophy: apply minimal perturbation only when the natural orbit cooperates.

The choice among these methods depends on context. If the map can be modified offline (static control is acceptable), $\alpha(x)$ provides superlinear convergence and simple implementation. If only real-time feedback is available, Pyragas with correct gain provides complete basin coverage with modest feedback magnitude. OGY is appropriate when minimal intervention is essential and the orbit naturally visits the target neighborhood.

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